



SECURITY · PRIVACY · QA · DATA GOVERNANCE

Every arrow is a decision.

Every product you use moves data along arrows: from your finger to a chip, from the chip to a server, from the server to someone else. Security asks who could get at it. Privacy asks whether it should move at all. QA asks how anyone outside would know it behaves as promised. Data governance asks who holds it, how long it lives, and whether it can really be deleted. Here are eight everyday products, as Apple, Google and others document them, through all four.

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30 MIN ESSAY · 52 WITH EVERY TABLE 8 PRODUCTS · 8 COMPANIES · 138 CITED CLAIMS

01 FOUR LENSES

One arrow, four lenses.

Take the most ordinary arrow in modern life: you glance at your phone, and a website lets you in. Four disciplines look at that one arrow and ask four different questions.

SECURITY	PRIVACY	QA	DATA GOVERNANCE
Can the wrong actor get at it?	Should it exist, move, combine, persist or be inferred?	How would anyone outside know it works as promised?	Who holds it, how long does it live, and can it really go?
Could a look-alike site replay the login? Could someone who steals the cloud copy use the key?	Does your face leave the phone? Does the site learn anything that links you to other sites?	Which test proves a look-alike domain is refused? Who outside the company can check the claim?	Who can restore the key when every device is lost? When you delete it on the phone, does the site forget it too?

Most reviews use one lens. Security teams ask the first question, privacy teams the second, QA teams test what the specification says, and governance teams keep the register. The failures live in the gaps between them: a login that cannot be phished but can be recovered by anyone who answers a text message; a backup that is encrypted but readable by whoever holds the key; a setting that says “don’t save” with no test that shows nothing was saved.

This edition looks at eight products most people use every day, as three companies each document them, through all four lenses at once. Apple and Google appear in every row; the third column is whoever makes the other product people most often compare them with: Microsoft, Mozilla, Proton, Meta, Samsung or Amazon.

How to read the tables

DOCUMENTED The company says so in its own documentation.	SETTING It depends on a setting; the default is given where the company documents one.	LIMIT A limitation the company itself documents, in its own terms.	TEST Ours, not theirs: what a QA engineer should check before trusting the claim.
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This is not a ranking. There are no scores and no winners, and a longer column is not a better product. Products change month to month: every claim cites the company’s own page as reviewed on 30 September 2026, and [the method](#) lists what could not be confirmed.

02 THE WHOLE PICTURE

Eight products, one lens at a time.

Pick a lens. Each row is a product, each column a company, each cell the headline of what that company documents. The product name takes you to the detail.

SECURITY CAN THE WRONG ACTOR GET AT IT?

Product	Apple	Google	And one more
Sign-in	DOCUMENTED Synced end-to-end encrypted	DOCUMENTED Synced end-to-end encrypted	MICROSOFT DOCUMENTED Phishing-resistant by design
Browser	DOCUMENTED Browser escapes are paid for	DOCUMENTED Each site in its own process SETTING Safe Browsing has two levels	MOZILLA DOCUMENTED Sync is end-to-end encrypted
Mail	LIMIT Stored mail is not end-to-end encrypted	LIMIT Confidential mode is not encryption	PROTON DOCUMENTED End-to-end between Proton users
Messages	DOCUMENTED End-to-end, post-quantum	DOCUMENTED End-to-end when every condition holds DOCUMENTED Across iPhone and Android, in beta	META DOCUMENTED End-to-end, always on
Wallet & ID	DOCUMENTED A device number, not your card number DOCUMENTED The ID is bound to the phone	DOCUMENTED A virtual card number DOCUMENTED IDs stay on the device	SAMSUNG DOCUMENTED Tokens, and Knox underneath
Cloud backup	DOCUMENTED Standard protection by default	DOCUMENTED Device backups end-to-end encrypted DOCUMENTED Guesses limited in secure hardware	MICROSOFT DOCUMENTED Encrypted at rest, keys held by Microsoft SETTING Personal Vault adds a second check
AI assistant	DOCUMENTED Private Cloud Compute	DOCUMENTED Private AI Compute DOCUMENTED Gemini Nano on the device	MICROSOFT DOCUMENTED Recall is sealed on the PC
Home & voice	DOCUMENTED Commands encrypted to the accessory	DOCUMENTED Five years of updates	AMAZON DOCUMENTED Nothing sent before the wake word SETTING Ring offers end-to-end video, optionally

PRIVACY SHOULD IT EXIST, MOVE, COMBINE, PERSIST OR BE INFERRED?

Product	Apple	Google	And one more
<u>Sign-in</u>	DOCUMENTED A new key pair for every account	DOCUMENTED Biometrics stay on the device	MICROSOFT DOCUMENTED Windows Hello data does not roam
<u>Browser</u>	DOCUMENTED Fingerprinting limits in every tab SETTING Private Relay hides your IP address	SETTING Third-party cookies stay on DOCUMENTED The replacement plan was retired	MOZILLA DOCUMENTED One cookie jar per site, by default SETTING Standard blocks known trackers
<u>Mail</u>	SETTING Mail Privacy Protection DOCUMENTED A different address for every sender	DOCUMENTED Mail is not used for ads SETTING Images arrive through Google's proxy	PROTON LIMIT Subjects and addresses are not end-to-end SETTING Tracker protection
<u>Messages</u>	LIMIT Routing data remains	LIMIT Operational metadata remains LIMIT Chats with Gemini are not end-to-end	META LIMIT Metadata is collected DOCUMENTED AI requests in sealed hardware
<u>Wallet & ID</u>	DOCUMENTED Apple does not see where the ID is shown DOCUMENTED Card payments are not tied back to you	DOCUMENTED Prove your age with a zero-knowledge proof SETTING Transaction data, and a personalisation switch	SAMSUNG DOCUMENTED Transaction and usage data are collected DOCUMENTED A standard mobile ID
<u>Cloud backup</u>	SETTING Advanced Data Protection is opt-in LIMIT Not offered to new UK users	LIMIT Photos are not end-to-end encrypted SETTING Customer-held keys for Workspace	MICROSOFT LIMIT A vault is a gate, not end-to-end encryption
<u>AI assistant</u>	DOCUMENTED Used for the request, then not kept SETTING ChatGPT is opt-in	SETTING Activity is kept by default	MICROSOFT SETTING Recall is opt-in and local SETTING Consumer Copilot training toggles
<u>Home & voice</u>	SETTING Audio kept only if you opt in	SETTING Recordings not saved by default	AMAZON LIMIT The local-processing option was removed SETTING Don't save, and a review switch

QA HOW WOULD ANYONE OUTSIDE KNOW IT WORKS AS PROMISED?

Product	Apple	Google	And one more
<u>Sign-in</u>	DOCUMENTED Published design, paid bounty TEST Delete at the site, then look on the phone	DOCUMENTED Open code, paid bounty TEST Forget the PIN on a new laptop	MICROSOFT DOCUMENTED Certified, and in the identity bounty TEST Delete in one place, check the other two
<u>Browser</u>	DOCUMENTED The engine is open source TEST Read the canvas twice	DOCUMENTED Open code, paid bounty TEST Diff what leaves	MOZILLA DOCUMENTED Open source, with a client bounty TEST Two sites, one iframe
<u>Mail</u>	DOCUMENTED Closed app, paid bounty TEST Fire a pixel you control	DOCUMENTED Covered by Google's rewards TEST When, and from where, does the pixel load?	PROTON DOCUMENTED Open-source, audited apps TEST Read what the server can read
<u>Messages</u>	SETTING Check the keys yourself TEST Add a device and watch for the alert	DOCUMENTED Published protocol paper TEST A matrix, not a single test	META DOCUMENTED Key transparency and a bounty TEST Reinstall and verify
<u>Wallet & ID</u>	DOCUMENTED Documented components, paid bounty TEST Ask only for "over 21"	DOCUMENTED The cryptography is public TEST Two presentations, unlinkable?	SAMSUNG DOCUMENTED A mobile rewards programme TEST Trip the integrity check
<u>Cloud backup</u>	DOCUMENTED Key design published TEST Try to enrol without a way back	DOCUMENTED A public third-party audit TEST Wrong PIN on a new phone	MICROSOFT DOCUMENTED OneDrive is in the bounty TEST Day 29 and day 31
<u>AI assistant</u>	DOCUMENTED Verifiable by outsiders SETTING A report of what went to the cloud TEST Replay and reconcile	DOCUMENTED AI is in the bounty TEST Turn it off and check the export	MICROSOFT DOCUMENTED Tested, and in the AI bounty TEST Feed it a fake card number
<u>Home & voice</u>	DOCUMENTED Camera video end-to-end encrypted TEST Opt out, then ask for everything	DOCUMENTED Third-party assessments published TEST Default settings, then Takeout	AMAZON DOCUMENTED Devices are in a bounty LIMIT A regulator's record TEST Transcripts too

DATA GOVERNANCE

WHO HOLDS IT, HOW LONG DOES IT LIVE, AND CAN IT REALLY GO?

Product	Apple	Google	And one more
<u>Sign-in</u>	DOCUMENTED Recovery contact or recovery key DOCUMENTED Passkeys can move between managers	LIMIT Forgetting the PIN can cost every passkey LIMIT Removing it from the account is only half	MICROSOFT DOCUMENTED Passwordless by default for new accounts LIMIT Three places to delete
<u>Browser</u>	LIMIT Bookmarks follow your iCloud setting	SETTING Administrators control sign-in and sync	MOZILLA LIMIT Forget the password, lose the synced data DOCUMENTED Terms rewritten in public
<u>Mail</u>	DOCUMENTED A copy on request, requests counted	SETTING Customer-held keys for some Workspace editions DOCUMENTED A warrant for content, requests counted	PROTON DOCUMENTED Swiss orders, published DOCUMENTED Your mail, out
<u>Messages</u>	LIMIT The backup can hold the key LIMIT Not offered to new UK users	DOCUMENTED Backups tied to your screen lock	META SETTING Encrypted backups are opt-in DOCUMENTED Content is not produced
<u>Wallet & ID</u>	DOCUMENTED Enrolment data deleted after issue LIMIT US only	LIMIT Device-only means no export	SAMSUNG LIMIT A compromised phone loses Wallet for good DOCUMENTED Delete per device or per account
<u>Cloud backup</u>	LIMIT You hold the only recovery DOCUMENTED A copy on request	DOCUMENTED Takeout and a 30-day bin DOCUMENTED Inactive accounts can be deleted	MICROSOFT DOCUMENTED A 30-day recycle bin LIMIT Over quota for six months
<u>AI assistant</u>	DOCUMENTED Not used for training	SETTING 18 months by default LIMIT Reviewed chats stay longer DOCUMENTED Workspace data is not used for training	MICROSOFT SETTING Organisations can remove Recall DOCUMENTED Enterprise prompts are not training data
<u>Home & voice</u>	DOCUMENTED Ten days of video	SETTING Auto-delete after 18 months	AMAZON SETTING Four retention choices

Security across 8 products: 27 documented, 3 settings, 2 limits, 0 tests.

03 SIGN-IN

Who proves you are you?

A passkey replaces the password with a key pair: the private half stays in your devices, the website keeps the public half, and your face or PIN only unlocks the key. All three companies now sync passkeys between your devices. The questions move from “can someone steal the password?” to “who can restore the key when every device is lost?”



The highlighted hop is the arrow to watch. Compared: **Apple** Passkeys in iCloud Keychain · **Google** Passkeys in Google Password Manager · **Microsoft** Passkeys and Windows Hello.

SIGN-IN: WHAT EACH COMPANY DOCUMENTS, THROUGH FOUR LENSES

Lens	Apple Passkeys in iCloud Keychain	Google Passkeys in Google Password Manager	Microsoft Passkeys and Windows Hello
SECURITY Can the wrong actor get at it?	DOCUMENTED Synched end-to-end encrypted Apple documents that iCloud Keychain is end-to-end encrypted with keys Apple does not know, and that recovery is rate-limited against brute force even from a privileged position on its own cloud. Apple Support	DOCUMENTED Synched end-to-end encrypted Google documents that passkeys in Google Password Manager are end-to-end encrypted, and that adding a new device needs your screen lock or a Password Manager PIN. Google	DOCUMENTED Phishing-resistant by design Microsoft describes passkeys as a standards-based, phishing-resistant method and reports far higher sign-in success with passkeys than with passwords. Microsoft Security Blog
PRIVACY Should it exist, move, combine, persist or be inferred?	DOCUMENTED A new key pair for every account The site never receives the private key, and Face ID or Touch ID only authorises its use; the biometric is not sent anywhere. Apple Support · Apple Platform Security	DOCUMENTED Biometrics stay on the device Google states that fingerprint and face data stay on the device and are never shared with Google. Google Account	DOCUMENTED Windows Hello data does not roam Microsoft documents that the Windows Hello PIN and biometric data stay on the device, protected by the TPM, and are not synced. Microsoft Learn

Lens	Apple Passkeys in iCloud Keychain	Google Passkeys in Google Password Manager	Microsoft Passkeys and Windows Hello
QA How would anyone outside know it works as promised?	DOCUMENTED Published design, paid bounty The Apple Platform Security guide describes keychain sync, and the Apple Security Bounty pays for breaking it. Apple · Apple Security Research TEST Delete at the site, then look on the phone Remove a passkey at a test site and confirm the Passwords app stops offering it once the site signals the change. Confirm a look-alike domain is never offered the credential.	DOCUMENTED Open code, paid bounty Chromium is public source, and Chrome security bugs are paid through Google's vulnerability reward programme. Chromium TEST Forget the PIN on a new laptop Enter a wrong Password Manager PIN until it locks, then follow the documented reset and confirm what it deletes.	DOCUMENTED Certified, and in the identity bounty Windows Hello received FIDO2 certification in 2019, and Microsoft's identity bounty covers Microsoft accounts. FIDO Alliance · Microsoft Security Response Center TEST Delete in one place, check the other two Remove a passkey in Windows settings and confirm the site still lists it until it is removed from the account too.
DATA GOVERNANCE Who holds it, how long does it live, and can it really go?	DOCUMENTED Recovery contact or recovery key Account recovery uses a trusted Recovery Contact or an optional 28-character recovery key. Deleted passkeys can be recovered for 30 days. Apple Platform Security · Apple Support DOCUMENTED Passkeys can move between managers In the 2025 releases a user can move passkeys to another credential manager, gated by Face ID or Touch ID and using a FIDO Alliance format, with no unencrypted file. Apple Developer	LIMIT Forgetting the PIN can cost every passkey If the PIN cannot be recovered from any device, the documented last resort is resetting passkeys, which deletes them. Google Chrome Help LIMIT Removing it from the account is only half Google's passkey guidance notes that removing a passkey from the account removes the server's public key; the private key stays in the credential manager until the user deletes it there. Google for Developers	DOCUMENTED Passwordless by default for new accounts Since May 2025, new Microsoft accounts are created without a password by default. Microsoft Security Blog LIMIT Three places to delete Microsoft documents separate removal on the account, on the device and in the syncing provider. Microsoft Support

All four lenses for Apple, Google and Microsoft: 12 documented, 0 settings, 3 limits, 3 tests.

WHERE THEY AGREE

All three keep the biometric on the device, give each site its own public key, and sync the private keys end-to-end encrypted. The phishing arrow is closed the same way everywhere.

WHERE THEY DIFFER

What happens when you lose everything differs: Apple offers a recovery contact or key, Google's documented fallback is a reset that deletes passkeys, and Microsoft splits deletion across the account, the device and the sync provider.

THE ARROW TO WATCH

The recovery path. It is where an end-to-end encrypted key is restored or lost for good, and where deleting on the phone and deleting at the site fall out of step.

04 BROWSER

Who else is on the page?

The page you see is one arrow. The scripts, cookies and requests you do not see are dozens more, each going to a party you did not visit. A browser decides by default which of those arrows fire, and each of these three decides differently.



The highlighted hop is the arrow to watch. Compared: **Apple** Safari · **Google** Chrome · **Mozilla** Firefox.

BROWSER: WHAT EACH COMPANY DOCUMENTS, THROUGH FOUR LENSES

Lens	Apple Safari	Google Chrome	Mozilla Firefox
SECURITY Can the wrong actor get at it?	DOCUMENTED Browser escapes are paid for The Apple Security Bounty pays for WebKit sandbox escapes. Apple Security Research	DOCUMENTED Each site in its own process Chrome's Site Isolation puts sites in separate sandboxed renderer processes. Chromium SETTING Safe Browsing has two levels Standard protection checks a local list and sends a partial, obfuscated URL through a privacy server. Enhanced protection is opt-in and sends more, including URLs and downloads, to Google. Google Chrome Help	DOCUMENTED Sync is end-to-end encrypted Mozilla documents that Firefox Sync data is encrypted before it leaves the browser and the password that unlocks it is never sent to Mozilla. Mozilla Support

Lens	Apple Safari	Google Chrome	Mozilla Firefox
PRIVACY Should it exist, move, combine, persist or be inferred?	DOCUMENTED Fingerprinting limits in every tab In Safari 26, Advanced Fingerprinting Protection is on by default for all browsing, restricting known fingerprinting scripts' access to APIs such as canvas, audio and screen. WebKit SETTING Private Relay hides your IP address With iCloud+, Private Relay sends Safari traffic through two relays so no single party sees both your IP address and the site. It is off unless you subscribe and turn it on. Apple Support	SETTING Third-party cookies stay on Chrome allows third-party cookies by default in normal browsing and blocks them by default in Incognito. Chrome Enterprise DOCUMENTED The replacement plan was retired Google said in April 2025 it would keep third-party cookies, and in October 2025 retired most Privacy Sandbox APIs, keeping CHIPS, FedCM and Private State Tokens. Privacy Sandbox	DOCUMENTED One cookie jar per site, by default Total Cookie Protection has been on by default for desktop users since June 2022. Mozilla SETTING Standard blocks known trackers Enhanced Tracking Protection "Standard" blocks social-media trackers, cross-site cookies, cryptominers and known fingerprinters; newer fingerprinting defences started in private windows and Strict mode. Mozilla Support · Mozilla
QA How would anyone outside know it works as promised?	DOCUMENTED The engine is open source WebKit is open source and its changes are published. WebKit TEST Read the canvas twice In a normal (not private) tab, run a fingerprinting test page and compare canvas and audio readbacks across reloads.	DOCUMENTED Open code, paid bounty Chromium is public source, with a vulnerability reward programme. Chromium TEST Diff what leaves Capture traffic with Safe Browsing on Standard, then Enhanced, and compare what reaches Google with the documented lists.	DOCUMENTED Open source, with a client bounty Firefox is open source and Mozilla's client bug bounty pays for critical and high-severity flaws. Mozilla TEST Two sites, one iframe Embed the same third-party frame in two sites, write a cookie in one and confirm the other cannot read it in Standard mode.

Lens	Apple Safari	Google Chrome	Mozilla Firefox
DATA GOVERNANCE Who holds it, how long does it live, and can it really go?	LIMIT Bookmarks follow your iCloud setting Apple lists Safari Tab Groups as end-to-end encrypted; bookmarks are end-to-end encrypted only with Advanced Data Protection on. Apple Support	SETTING Administrators control sign-in and sync Enterprise policies can force or disable browser sign-in and restrict which data types sync. Chrome Enterprise	LIMIT Forget the password, lose the synced data Because Mozilla never has the key, it cannot recover synced data if you forget your account password. Mozilla Support DOCUMENTED Terms rewritten in public In 2025 Mozilla introduced terms of use for Firefox and revised them after criticism, tying its rights to the privacy notice. Mozilla

All four lenses for Apple, Google and Mozilla: 10 documented, 5 settings, 2 limits, 3 tests.

WHERE THEY AGREE

All three isolate or partition something by default and run public bug bounties. All three let organisations manage the browser by policy.

WHERE THEY DIFFER

Defaults diverge: Safari restricts known fingerprinting scripts in every tab, Firefox partitions cookies per site, and Chrome keeps third-party cookies on in normal browsing after retiring its replacement plan.

THE ARROW TO WATCH

Third-party requests in a normal tab. That is where the default decides who else learns what you read.

05 MAIL

Does the email read you back?

An email can carry a one-pixel image. When your app fetches it, the sender learns that you opened the message, when, and from which IP address. Mail providers now step between the pixel and the reader, but none end-to-end encrypts the subject line or the envelope of ordinary email.



The highlighted hop is the arrow to watch. Compared: **Apple** Mail and iCloud Mail · **Google** Gmail · **Proton** Proton Mail.

MAIL: WHAT EACH COMPANY DOCUMENTS, THROUGH FOUR LENSES

Lens	Apple Mail and iCloud Mail	Google Gmail	Proton Proton Mail
SECURITY Can the wrong actor get at it?	LIMIT Stored mail is not end-to-end encrypted Apple lists iCloud Mail among the categories that stay outside end-to-end encryption even with Advanced Data Protection, because mail must interoperate with the global email system. Apple Support	LIMIT Confidential mode is not encryption Google documents that confidential mode blocks forwarding, copying, printing and downloading and lets the sender revoke access, but cannot stop screenshots or photos. Gmail Help	DOCUMENTED End-to-end between Proton users Proton documents end-to-end encryption for bodies and attachments between Proton users and for password-protected mail; mail from other providers arrives over TLS and is then stored with zero-access encryption. Proton Support
PRIVACY Should it exist, move, combine, persist or be inferred?	SETTING Mail Privacy Protection With Protect Mail Activity on, your IP address is hidden from senders and remote content is downloaded privately in the background when a message arrives, not when you open it. Apple Support DOCUMENTED A different address for every sender Hide My Email in iCloud+ creates unique, random addresses that forward to your inbox. Apple Support	DOCUMENTED Mail is not used for ads Google stopped using consumer Gmail content for ads personalisation in 2017. Google SETTING Images arrive through Google's proxy Gmail serves message images through Google's own proxy servers; showing external images automatically is a setting. Gmail Help	LIMIT Subjects and addresses are not end-to-end Proton states that subject lines and sender and recipient addresses are encrypted, but not end-to-end, for compatibility with email standards. Proton Support SETTING Tracker protection Proton blocks known tracking pixels, loads remote images through its proxy and cleans tracking links on the web client. Proton Support

Lens	Apple Mail and iCloud Mail	Google Gmail	Proton Proton Mail
QA How would anyone outside know it works as promised?	DOCUMENTED Closed app, paid bounty The Apple Security Bounty includes unauthorised access to iCloud account data. Apple Security Research	DOCUMENTED Covered by Google's rewards Google's vulnerability reward programme covers issues that substantially affect the confidentiality or integrity of user data. Google Bug Hunters	DOCUMENTED Open-source, audited apps Proton publishes its app source code and independent audit reports, and runs a public bug bounty. Proton · Proton
	TEST Fire a pixel you control Log the image fetch: with protection on, it should come at delivery from a relay address, never at opening from the reader's own IP.	TEST When, and from where, does the pixel load? Log the fetch with external images on and off; revoke a confidential-mode message and confirm it is unreadable on web and mobile.	TEST Read what the server can read Send from the open-source client to an outside address and confirm the server sees the subject and addresses, as documented, but never a Proton-to-Proton body.
DATA GOVERNANCE Who holds it, how long does it live, and can it really go?	DOCUMENTED A copy on request, requests counted Apple's Data and Privacy portal returns a copy of account data, and a twice-yearly Transparency Report counts government requests. Apple Support · Apple	SETTING Customer-held keys for some Workspace editions Client-side encryption encrypts body and attachments with keys the organisation holds; headers, subject, recipients and timestamps are not encrypted. It is off by default. Gmail Help	DOCUMENTED Swiss orders, published Proton's transparency report counts the legally binding orders it receives under Swiss law. Proton
		DOCUMENTED A warrant for content, requests counted Google says it requires a search warrant for Gmail content in criminal cases and publishes request statistics every six months. Google Transparency Report Help	DOCUMENTED Your mail, out A free, open-source export tool downloads decrypted mail with its metadata. Proton Support

All four lenses for Apple, Google and Proton: 10 documented, 4 settings, 3 limits, 3 tests.

WHERE THEY AGREE

All three put something between the sender's remote content and the reader — a relay, a proxy or a blocker. None end-to-end encrypts the subject line or the envelope of ordinary email.

WHERE THEY DIFFER

Apple and Google hold the keys to stored mail (Workspace client-side encryption aside); Proton cannot read stored mail. Proton's apps are open source and audited; Apple's and Google's are closed and checked through bounties.

THE ARROW TO WATCH

The pixel fetch. When it happens and whose IP address it comes from can be tested directly, and it is the whole of the privacy claim.

06 MESSAGES

Who else can read the conversation?

All three encrypt the conversation end to end. The differences live at the edges: whether the protection can switch off, what the servers still see, and who holds the key to the backup.



The highlighted hop is the arrow to watch. Compared: **Apple** iMessage · **Google** Google Messages (RCS) · **Meta** WhatsApp.

MESSAGES: WHAT EACH COMPANY DOCUMENTS, THROUGH FOUR LENSES

Lens	Apple iMessage	Google Google Messages (RCS)	Meta WhatsApp
SECURITY Can the wrong actor get at it?	DOCUMENTED End-to-end, post-quantum Apple documents that iMessage content and attachments are end-to-end encrypted and not stored by Apple, and has rolled out the post-quantum PQ3 protocol since iOS 17.4. Apple Platform Security · Apple Security Research	DOCUMENTED End-to-end when every condition holds RCS chats in Google Messages are end-to-end encrypted, one-to-one and in groups, when everyone uses RCS; SMS and MMS never are. A lock icon marks encrypted chats. Google Messages Help DOCUMENTED Across iPhone and Android, in beta End-to-end encrypted RCS between Android and iPhone began rolling out in May 2026, needing current apps and carrier support on both ends. Google · Apple Newsroom	DOCUMENTED End-to-end, always on WhatsApp documents end-to-end encryption for messages, media, voice notes and calls, and that there is no way to turn it off. WhatsApp Blog

Lens	Apple iMessage	Google Google Messages (RCS)	Meta WhatsApp
PRIVACY Should it exist, move, combine, persist or be inferred?	LIMIT Routing data remains Timestamps and push-routing metadata are not end-to-end encrypted; Apple's legal guidelines say lookup logs, without content, are kept up to 25 days. Apple Platform Security · Apple	LIMIT Operational metadata remains Google documents that its servers can still use sender and recipient numbers, timestamps, IP or connection information and carriers. Google Messages Help LIMIT Chats with Gemini are not end-to-end Conversations with Gemini inside Messages are not end-to-end encrypted. Gemini Help	LIMIT Metadata is collected WhatsApp's privacy policy lists usage and log information — activity, frequency, online status — and connection information such as IP address and device identifiers. WhatsApp DOCUMENTED AI requests in sealed hardware Meta documents Private Processing: AI requests go encrypted into trusted execution environments that it says Meta cannot access, and nothing is stored. Engineering at Meta
QA How would anyone outside know it works as promised?	SETTING Check the keys yourself Contact Key Verification, when turned on, alerts you if a key the servers present does not match and lets two people compare codes. Apple Support TEST Add a device and watch for the alert With verification on for two test accounts, add a new device to one and confirm the other is warned.	DOCUMENTED Published protocol paper Google publishes a technical paper describing its end-to-end encryption; the app itself is closed. Google TEST A matrix, not a single test Try Android and iPhone, supported and unsupported carriers, one-to-one and group; the lock must appear only when every condition holds, and a fall-back to SMS must be visible every time.	DOCUMENTED Key transparency and a bounty WhatsApp publishes key transparency through an open-source auditable key directory, and Meta's bug bounty covers WhatsApp and Private Processing. Engineering at Meta · Meta Bug Bounty TEST Reinstall and verify Reinstall one test account and confirm the security-code change is shown to the other side and the key-transparency proof verifies.

Lens	Apple iMessage	Google Google Messages (RCS)	Meta WhatsApp
DATA GOVERNANCE Who holds it, how long does it live, and can it really go?	LIMIT The backup can hold the key With iCloud Backup on and Advanced Data Protection off, the backup includes a copy of the Messages in iCloud key, protected by keys Apple holds. Advanced Data Protection is off by default. Apple Support LIMIT Not offered to new UK users Since February 2025 Apple cannot offer Advanced Data Protection to new users in the United Kingdom. Apple Support	DOCUMENTED Backups tied to your screen lock Android backup end-to-end encrypts some data with the screen-lock PIN, pattern or password. Android Help	SETTING Encrypted backups are opt-in End-to-end encrypted backups are optional, secured by a password or a 64-digit key; if the key is lost, WhatsApp cannot restore the backup. Engineering at Meta DOCUMENTED Content is not produced WhatsApp says it cannot and does not produce message content to governments; Meta publishes request data twice a year. WhatsApp Help Center

All four lenses for Apple, Google and Meta: 9 documented, 2 settings, 6 limits, 3 tests.

WHERE THEY AGREE

All three end-to-end encrypt the content of native conversations, and all three document that servers still see routing metadata. Each offers some way to check keys.

WHERE THEY DIFFER

WhatsApp's encryption cannot be switched off; Google Messages depends on RCS, the other person's phone and the carrier. On backups, Apple's and WhatsApp's strongest options are opt-in, while Android ties backup encryption to the screen lock.

THE ARROW TO WATCH

The backup. In transit the message is encrypted by default everywhere; the backup copy is where defaults decide who holds the key.

07 WALLET & ID

What does the tap reveal?

Tapping to pay sends a stand-in for your card number. Showing a digital ID can prove you are over 21 without your name or birth date. The token protects you from the merchant; what the wallet maker itself keeps is a separate question.



The highlighted hop is the arrow to watch. Compared: **Apple** Apple Wallet and Apple Pay · **Google** Google Wallet · **Samsung** Samsung Wallet.

WALLET & ID: WHAT EACH COMPANY DOCUMENTS, THROUGH FOUR LENSES

Lens	Apple Apple Wallet and Apple Pay	Google Google Wallet	Samsung Samsung Wallet
SECURITY Can the wrong actor get at it?	DOCUMENTED A device number, not your card number Apple documents that the issuer creates a Device Account Number, kept in the Secure Element, never stored on Apple’s servers or backed up to iCloud; each payment adds a one-time code. Apple Support DOCUMENTED The ID is bound to the phone The private key for an ID in Wallet stays in the Secure Element, and each data element is signed. Apple Platform Security	DOCUMENTED A virtual card number Google documents virtual card numbers that stand in for the real card online and in apps. Google Pay Help DOCUMENTED IDs stay on the device ID passes and state IDs are stored encrypted on the device, not in the Google Account, and nothing is shared without device authentication. Google Wallet Help	DOCUMENTED Tokens, and Knox underneath Samsung documents that tokenisation replaces the card number and that Knox keeps payment and biometric data in a hardware-isolated environment. Samsung US Support

Lens	Apple Apple Wallet and Apple Pay	Google Google Wallet	Samsung Samsung Wallet
PRIVACY Should it exist, move, combine, persist or be inferred?	DOCUMENTED Apple does not see where the ID is shown Apple states it cannot see when or where you present your Digital ID or what was shared; only the requested fields leave, after Face ID or Touch ID. Apple Newsroom · Apple Support DOCUMENTED Card payments are not tied back to you Apple documents that it does not keep transaction information that can be tied back to you for card payments. Apple Support	DOCUMENTED Prove your age with a zero-knowledge proof Google's age assurance proves "over 18" without revealing a birth date, and it open-sourced the proof libraries in 2025. Google SETTING Transaction data, and a personalisation switch The Wallet privacy notice documents collecting transaction data; a Wallet personalisation setting controls its use across Google. Google · Google Wallet Help	DOCUMENTED Transaction and usage data are collected Samsung's Wallet privacy notice lists payment attempts, merchants and amounts, and how and when Wallet is used. Samsung US DOCUMENTED A standard mobile ID Samsung's digital ID follows the ISO mobile driving licence standard, is stored on the device and is released only after fingerprint or PIN. Samsung US
QA How would anyone outside know it works as promised?	DOCUMENTED Documented components, paid bounty The Apple Platform Security guide documents Apple Pay components; the Security Bounty pays for breaking them. Apple Platform Security · Apple Security Research TEST Ask only for "over 21" Present an ID to a test ISO 18013-5 reader requesting only age; confirm no birth date, name or address comes back.	DOCUMENTED The cryptography is public The zero-knowledge proof library is open source and was presented to the IETF research group, so the maths itself can be reviewed. Google · IETF 124 TEST Two presentations, unlinkable? Request age-over-18 twice from the same wallet; confirm the verifier gets a valid proof, no birth date, and bytes that cannot be linked.	DOCUMENTED A mobile rewards programme Samsung runs a mobile security rewards programme with Knox in scope. Samsung Mobile Security TEST Trip the integrity check On a test device, unlock the bootloader and confirm Wallet refuses to provision or pay; wipe Wallet remotely and confirm the issuer shows the tokens deactivated.

Lens	Apple Apple Wallet and Apple Pay	Google Google Wallet	Samsung Samsung Wallet
DATA GOVERNANCE Who holds it, how long does it live, and can it really go?	DOCUMENTED Enrolment data deleted after issue Apple documents that passport chip data is deleted from its servers right after issuance, and the selfie and movement video shortly after approval. Apple Legal	LIMIT Device-only means no export Because the ID is saved on the device and not in the account, Google Takeout cannot export it. Google Wallet Help	LIMIT A compromised phone loses Wallet for good Knox permanently disables Samsung Wallet on a device it detects as compromised. Samsung US Support
	LIMIT US only The Digital ID from a US passport works only in the US and does not replace a passport for travel. Apple Newsroom		DOCUMENTED Delete per device or per account Wallet data can be removed per device or account-wide through Samsung's privacy portal. Samsung US

All four lenses for Apple, Google and Samsung: 15 documented, 1 setting, 3 limits, 3 tests.

WHERE THEY AGREE

All three replace the card number with a device-bound token, gate payment and ID behind a biometric or PIN, and keep ID credentials on the device.

WHERE THEY DIFFER

Apple states it cannot see where an ID is shown and does not tie card transactions to you; Google and Samsung document collecting transaction details. Google alone has published its age-proof cryptography.

THE ARROW TO WATCH

Transaction data flowing from the wallet to the wallet maker. The token hides the card from the merchant; it says nothing about what the platform keeps.

08 CLOUD BACKUP

Who holds the key to your photos?

Every backup is encrypted. The question that separates them is who holds the key: the provider, so it can recover your account and scan the content, or only your devices, so nobody else can — including the provider, and including anyone who can compel it.



The highlighted hop is the arrow to watch. Compared: **Apple** iCloud · **Google** Android backup, Google Photos and Drive · **Microsoft** OneDrive.

CLOUD BACKUP: WHAT EACH COMPANY DOCUMENTS, THROUGH FOUR LENSES

Lens	Apple iCloud	Google Android backup, Google Photos and Drive	Microsoft OneDrive
SECURITY Can the wrong actor get at it?	DOCUMENTED Standard protection by default Apple documents that under Standard Data Protection it holds the keys for most categories so it can help with recovery; some categories, such as Keychain and Health, are always end-to-end encrypted. Apple Support	DOCUMENTED Device backups end-to-end encrypted Google documents that some Android backup data is end-to-end encrypted with the screen-lock PIN, pattern or password, and recommends a real lock rather than swipe. Android Help DOCUMENTED Guesses limited in secure hardware Google described backup keys wrapped by the lock-screen secret and held in Titan chips that rate-limit guesses. Google Security Blog	DOCUMENTED Encrypted at rest, keys held by Microsoft Microsoft documents BitLocker disk encryption and a unique key per file at rest, plus TLS in transit; Microsoft manages the keys. Microsoft Support SETTING Personal Vault adds a second check Personal Vault asks for a second verification step and locks after inactivity. Microsoft Support

Lens	Apple iCloud	Google Android backup, Google Photos and Drive	Microsoft OneDrive
PRIVACY Should it exist, move, combine, persist or be inferred?	SETTING Advanced Data Protection is opt-in Turning on Advanced Data Protection extends end-to-end encryption to Backup, Drive, Photos, Notes and more; Mail, Contacts and Calendar stay out because they must work with global standards. Apple Platform Security LIMIT Not offered to new UK users Since February 2025 Apple cannot offer Advanced Data Protection to new users in the United Kingdom. Apple Support	LIMIT Photos are not end-to-end encrypted Google documents that Google Photos content is not encrypted with the screen lock: it is encrypted in transit and at rest under keys Google controls. Android Help SETTING Customer-held keys for Workspace Workspace, not consumer accounts, offers client-side encryption with keys the organisation holds. Google Workspace Admin Help	LIMIT A vault is a gate, not end-to-end encryption Personal Vault files are encrypted in the cloud like other OneDrive files, under keys Microsoft holds. Microsoft Support · Microsoft Support
QA How would anyone outside know it works as promised?	DOCUMENTED Key design published The Platform Security guide documents iCloud and Advanced Data Protection keys; the Security Bounty covers unauthorised access to iCloud data. Apple Platform Security · Apple Security Research TEST Try to enrol without a way back Remove every recovery contact and key and confirm enrolment in Advanced Data Protection is blocked.	DOCUMENTED A public third-party audit NCC Group's 2018 audit of the backup design is public. NCC Group TEST Wrong PIN on a new phone Restore to a new device, enter a wrong lock-screen secret repeatedly and confirm app data locks out while Photos still syncs — showing where the end-to-end boundary sits.	DOCUMENTED OneDrive is in the bounty Microsoft's 365 bounty lists OneDrive domains in scope. Microsoft Security Response Center TEST Day 29 and day 31 Delete a file and confirm it restores from the recycle bin until day 30 and not after; lock the vault and confirm a direct link cannot read it.

Lens	Apple iCloud	Google Android backup, Google Photos and Drive	Microsoft OneDrive
DATA GOVERNANCE Who holds it, how long does it live, and can it really go?	LIMIT You hold the only recovery Under Advanced Data Protection Apple holds no recovery keys; you need a recovery contact or key, or risk permanent loss. Apple Platform Security DOCUMENTED A copy on request Apple's Data and Privacy portal returns account data and iCloud content after verifying identity. Apple Support	DOCUMENTED Takeout and a 30-day bin Google Takeout exports your data; Drive trash is deleted automatically after 30 days. Google Account Help · Google Drive Help DOCUMENTED Inactive accounts can be deleted Personal accounts inactive for two years may be deleted, including Drive and Photos content. Google	DOCUMENTED A 30-day recycle bin Recycle-bin items in personal accounts are deleted automatically 30 days after they are put there. Microsoft Support LIMIT Over quota for six months An account over its storage quota becomes read-only, and after six months its files may be deleted for good. Microsoft Support

All four lenses for Apple, Google and Microsoft: 11 documented, 3 settings, 5 limits, 3 tests.

WHERE THEY AGREE

All three encrypt in transit and at rest under provider-held keys by default, which lets the provider run recovery, search and abuse scanning. All three offer export and a 30-day bin.

WHERE THEY DIFFER

Only Apple offers consumers an end-to-end mode for photos and files, opt-in and not for new UK users. Google end-to-end encrypts device backups by default but not Photos or Drive. Microsoft's vault adds a gate, not provider-blind encryption.

THE ARROW TO WATCH

Key custody. Whoever holds the key — provider, user or recovery contact — decides who can be compelled to open it.

09 AI ASSISTANT

Where does the question go?

An assistant is useful because it reads across your mail, messages, photos and calendar. Where that reading happens — on the phone, in a sealed server, or in an ordinary cloud service — decides who else could see the request and how long it lives.



The highlighted hop is the arrow to watch. Compared: **Apple** Apple Intelligence · **Google** Gemini · **Microsoft** Copilot and Recall.

AI ASSISTANT: WHAT EACH COMPANY DOCUMENTS, THROUGH FOUR LENSES

Lens	Apple Apple Intelligence	Google Gemini	Microsoft Copilot and Recall
SECURITY Can the wrong actor get at it?	DOCUMENTED Private Cloud Compute Apple documents that the device encrypts each request to the keys of attested Private Cloud Compute nodes, so only a validated node can read it, and that there is no privileged runtime access. Apple Security Research	DOCUMENTED Private AI Compute Google announced Private AI Compute in November 2025: Gemini runs in attested, encrypted enclaves that Google says are inaccessible even to Google. Google DOCUMENTED Gemini Nano on the device Gemini Nano runs on the phone through a system service that Google says has no direct internet access. Android Developers Blog	DOCUMENTED Recall is sealed on the PC Microsoft documents that Recall encrypts snapshots and its index with TPM-protected keys usable only inside a virtualisation-based enclave, behind Windows Hello. Windows Experience Blog

Lens	Apple Apple Intelligence	Google Gemini	Microsoft Copilot and Recall
PRIVACY Should it exist, move, combine, persist or be inferred?	DOCUMENTED Used for the request, then not kept Apple says Private Cloud Compute uses personal data only to fulfil the request and never stores it. Apple Security Research · Apple Support SETTING ChatGPT is opt-in The ChatGPT extension is off unless enabled; signed out, the IP address is hidden and OpenAI must not store or train on requests; signed in, OpenAI's policies apply. Apple Legal	SETTING Activity is kept by default Keep Activity is on by default for adults; while on, chats can be reviewed by people and used to improve Google's AI. With it off, chats are still kept for up to 72 hours. Gemini Help	SETTING Recall is opt-in and local Recall is off unless the user turns it on, snapshots stay on the device, and a sensitive-information filter is on by default. Windows Experience Blog · Microsoft Support SETTING Consumer Copilot training toggles Consumer Copilot has separate opt-out toggles for training on text and voice conversations. Microsoft Support
QA How would anyone outside know it works as promised?	DOCUMENTED Verifiable by outsiders Apple publishes measurements of production software to a transparency log, provides a Virtual Research Environment, and pays bounties for breaking Private Cloud Compute. Apple Security Research SETTING A report of what went to the cloud Settings can export an Apple Intelligence Report listing the requests sent to Private Cloud Compute. Apple Support TEST Replay and reconcile Export the report, replay a fixed set of prompts and check that every cloud request matches a published software measurement.	DOCUMENTED AI is in the bounty Gemini is covered by Google's vulnerability reward programme. Google Bug Hunters TEST Turn it off and check the export With Keep Activity off, hold a chat, wait 72 hours and confirm a data export no longer contains it.	DOCUMENTED Tested, and in the AI bounty Microsoft says its offensive research team and a third party tested Recall; the Copilot bounty pays for AI flaws. Windows Experience Blog · Microsoft Security Response Center TEST Feed it a fake card number Open a page with test card and ID numbers, then search Recall; pass means no snapshot holds them.

Lens	Apple Apple Intelligence	Google Gemini	Microsoft Copilot and Recall
DATA GOVERNANCE Who holds it, how long does it live, and can it really go?	DOCUMENTED Not used for training Apple says it does not use users' private personal data or interactions to train its foundation models. Apple Machine Learning Research	SETTING 18 months by default Gemini activity auto-deletes after 18 months by default, adjustable or off. Gemini Help LIMIT Reviewed chats stay longer Chats already reviewed by people are disconnected from the account and kept up to three years, even after you delete your activity. Gemini Help DOCUMENTED Workspace data is not used for training For Workspace customers, Google says content is not reviewed by people or used to train models outside the domain without permission. Google Workspace	SETTING Organisations can remove Recall Recall is removed by default on managed commercial devices, and administrators can limit how long snapshots are kept. Microsoft Learn DOCUMENTED Enterprise prompts are not training data With enterprise data protection, Microsoft 365 Copilot prompts and responses are not used to train foundation models; they are retained for audit and eDiscovery. Microsoft Learn

All four lenses for Apple, Google and Microsoft: 11 documented, 7 settings, 1 limit, 3 tests.

WHERE THEY AGREE

All three split work between an on-device tier and a cloud tier, all three say work content is not used to train general models, and all three run bounties that cover their AI.

WHERE THEY DIFFER

Apple's cloud tier is documented as stateless and checkable through a public log. Google keeps consumer chats by default, with human review unless switched off. Microsoft's most intimate data, Recall, never leaves the PC and is opt-in.

THE ARROW TO WATCH

The arrow from device to cloud tier, and what is retained after the answer. "Verifiable" and "we promise" are different kinds of control.

10 HOME & VOICE

Who hears the kitchen?

A voice assistant listens for its wake word in a room where anyone can be speaking. What leaves the room, whether the audio is kept, and whether a person may listen to it are settings — and one of these companies removed a local-processing option in 2025.



The highlighted hop is the arrow to watch. Compared: **Apple** Siri and Apple Home · **Google** Google Assistant, Gemini for Home and Nest · **Amazon** Alexa, Echo and Ring.

HOME & VOICE: WHAT EACH COMPANY DOCUMENTS, THROUGH FOUR LENSES

Lens	Apple Siri and Apple Home	Google Google Assistant, Gemini for Home and Nest	Amazon Alexa, Echo and Ring
SECURITY Can the wrong actor get at it?	DOCUMENTED Commands encrypted to the accessory Apple documents that Home commands travel end-to-end encrypted from device through the home hub to the accessory, over HomeKit or Matter. Apple Platform Security	DOCUMENTED Five years of updates Google commits to automatic security updates for Nest devices for at least five years from launch, and verified boot on newer devices. Google Safety Center	DOCUMENTED Nothing sent before the wake word Amazon says Echo devices send nothing to the cloud until the wake word or button is detected, and requests are encrypted in transit. Amazon Help SETTING Ring offers end-to-end video, optionally Ring’s opt-in end-to-end encryption lets only enrolled phones decrypt video, and turns off some features. Ring Support

Lens	Apple Siri and Apple Home	Google Google Assistant, Gemini for Home and Nest	Amazon Alexa, Echo and Ring
PRIVACY Should it exist, move, combine, persist or be inferred?	SETTING Audio kept only if you opt in Apple says it does not retain Siri audio unless you opt in to Improve Siri & Dictation; opted-in samples are tied to a random identifier, not your Apple Account. Apple Newsroom · Apple Support	SETTING Recordings not saved by default Saving voice and audio recordings is off by default; a separate setting lets Google use recordings to improve its services. Google Nest Help · Google Home Help	LIMIT The local-processing option was removed From 28 March 2025, the “Do not send voice recordings” option on some Echo devices was withdrawn; affected users were moved to “Don’t save recordings”. TechCrunch SETTING Don’t save, and a review switch With “Don’t save recordings” on, audio is deleted after processing; whether recordings may be reviewed by people is a separate setting. Amazon Help · Amazon Help
QA How would anyone outside know it works as promised?	DOCUMENTED Camera video end-to-end encrypted HomeKit Secure Video is end-to-end encrypted, so Apple cannot view recordings. Apple Support TEST Opt out, then ask for everything With the improvement setting off, make 20 requests, then request a copy of your data and confirm no audio comes back.	DOCUMENTED Third-party assessments published Google publishes third-party security assessment results for Nest devices. Google TEST Default settings, then Takeout Leave audio saving off, use the assistant, then check My Activity and a Takeout export for audio. Pass means none.	DOCUMENTED Devices are in a bounty Echo and Alexa devices are in scope of Amazon’s vulnerability research programme. HackerOne LIMIT A regulator’s record In 2023 the FTC and DOJ charged that Amazon kept children’s Alexa recordings and undermined deletion requests; Amazon paid \$25 million to settle. U.S. Federal Trade Commission TEST Transcripts too Turn on “Don’t save recordings”, make requests, then export: no audio, and no transcripts derived from it.
DATA GOVERNANCE Who holds it, how long does it live, and can it really go?	DOCUMENTED Ten days of video HomeKit Secure Video keeps a rolling 10 days; opted-in Siri audio can be deleted. Apple Support · Apple Support	SETTING Auto-delete after 18 months Gemini for Home activity auto-deletes after 18 months by default, adjustable or off. Google Home Help	SETTING Four retention choices Recordings can be kept until deleted, for 18 months, for 3 months, or not saved. Amazon

All four lenses for Apple, Google and Amazon: 7 documented, 6 settings, 2 limits, 3 tests.

WHERE THEY AGREE

All three send at least some voice requests to the cloud and offer settings that limit whether recordings are kept or heard by people. All three publish security material for home devices.

WHERE THEY DIFFER

Apple keeps no Siri audio unless you opt in, and Google saves none by default. Amazon removed its local-processing option, so audio reaches the cloud and “don’t save” deletes it afterwards.

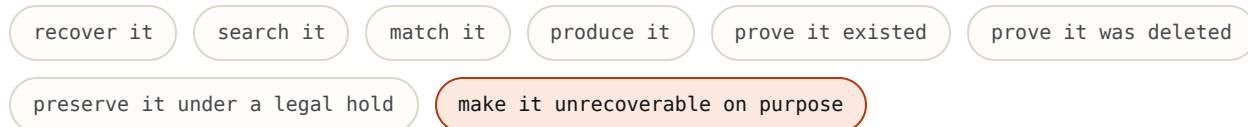
THE ARROW TO WATCH

The arrow from the speaker to the cloud, now that generative assistants need it — and whether deletion covers the transcripts made from the audio.

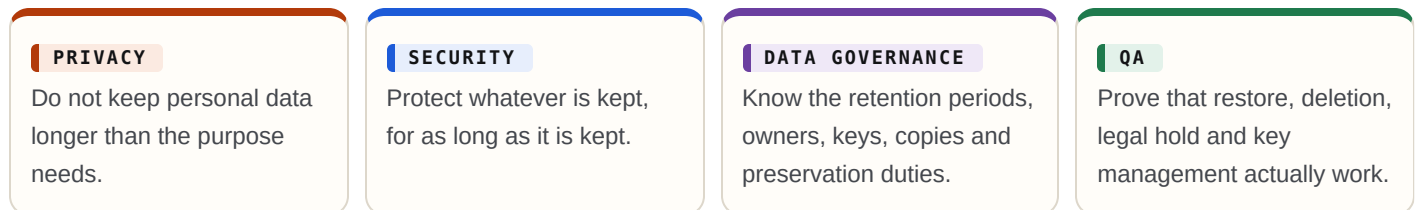
11 THE ARROW INTO THE FUTURE

What must still be possible years from now?

Every product above was read for where data goes, who can read it, how long it lives, whether it can be deleted, who holds the key and whether the promise can be checked. One arrow is missing from that list, because it points forward in time. A system also decides **what must still be possible with the data years from now**:



Those obligations pull in different directions, and each lens owns one of them.



Four ways to protect a value, four different futures

Take one value, an email address such as `dana@example.com`. There are four common ways to keep it out of plain sight. They look alike on a diagram and behave very differently ten years later.

Encryption

Kept: the value, scrambled. **The original comes back** for anyone who still holds the key.

- Who holds the key, and who could obtain it?
- Is the key kept as long as the data, and what happens when it is rotated?
- What if the key is destroyed on purpose?

Deleting the key can be equivalent to deleting the data. But if recovery is later required, a destroyed key leaves preserved bytes that nobody can read.

Tokenization

Kept: a stand-in, such as `tok_8f3a...`, in every system; the real value lives in a separate vault. **The original comes back** through the vault.

- Who controls the mapping?
- Does the vault follow the same retention rule as the tokens?
- Does a legal hold preserve the token only, or the mapping too? Can the token travel between systems without linking them?

The token may look harmless, but the vault decides whether identity can come back.

Hashing

Kept: a fixed fingerprint of the value. A cryptographic hash is designed to be one-way, and there is no key. **If the original is discarded, it cannot be recovered from the hash**, by anyone, under any order.

- If someone may legitimately need the original later, was an irreversible hash the right choice?
- How small is the set of possible inputs?

One-way does not necessarily mean unlinkable. When the possible inputs are guessable (a known list of email addresses, phone numbers, ZIP codes), anyone can hash each candidate and look for a match.

Keyed pseudonyms

Kept: a repeatable pseudonym made with a secret key (a keyed hash, such as HMAC), so analytics can count the same person without storing the address. **It cannot normally be reversed**, but a known candidate can be pseudonymised again and matched against old records, as long as the key, the normalisation rules (lower-case? trimmed?) and the algorithm version survive.

- Where is each historical key kept, and for how long?
- Are the normalisation rules and versions recorded alongside the data?

Key lifecycle can quietly become data lifecycle. Destroy the old key and the old records still exist, but can never be matched again.

When “delete” meets “preserve”

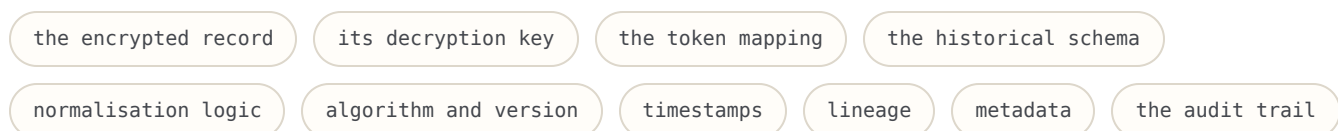
Most data follows a simple lifecycle:



An authorised legal or regulatory preservation requirement can bend it. A valid hold suspends normal deletion for the relevant records, for as long as the hold lasts:



The engineering question is not only “did we keep the file?” It is “did we keep enough context to make the file usable?” A record may depend on things stored somewhere else:



PRESERVED BYTES ARE NOT NECESSARILY PRESERVED EVIDENCE

A five-year-old archive that nobody can decrypt, interpret or match may technically exist while being operationally useless.

This describes engineering requirements, not legal advice: whether a hold applies, to what and for how long is for counsel to decide.

The future, through the four lenses

SECURITY

- Who can recover the protected data?
- Where are the keys, and who can use them?

PRIVACY

- Should the original remain recoverable at all?
- Would an irreversible transformation reduce the risk?

QA

- Can we restore an old archive?
- Can we reproduce a historical pseudonym?
- Does a legal hold really stop deletion, and does deletion resume when it is released?
- Do destroyed keys really make the data inaccessible?

DATA GOVERNANCE

- What is the retention rule, and what changes under a hold?
- Which copies, mappings, keys and derivatives does the hold include?

The products above show the same tension: an end-to-end protected backup can be [lost for good along with its recovery key](#), and every [30-day bin](#) is a copy with a clock.

12 EVERYDAY ARROWS

One morning, many arrows.

The products above were compared one at a time. A person meets most of them before ten in the morning, one ordinary action after another. Follow those actions instead of the products, and every one of them sets off the same chain:



Two of them, followed to the end:

Directions to a café

Identifiers: the account ID and the device ID. **Systems:** maps and the account service. **Derived:** the route, a location history, then a list of repeated places. **Inferred:** a possible commute, a likely home and workplace.

Only the route was needed to give directions. The history, the repeated places and the commute are further arrows, and each one is a decision.

“When is my flight?”

Identifier: the account. **Systems:** the assistant, mail and calendar. **Derived:** a reservation read from a confirmation email, and the schedule. **Inferred:** the travel itinerary.

Joined with the likely home from the maps trace, the itinerary also says when the home will be empty. Nobody asked that question.

Harmless alone, revealing together

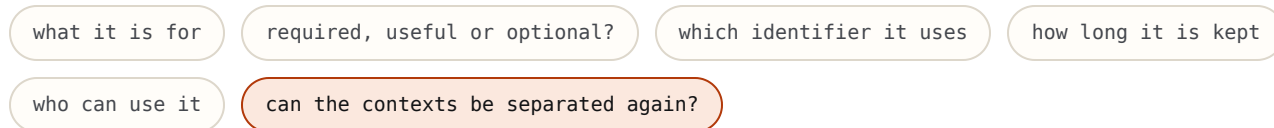
No single event in that morning is sensitive. Joined through one account ID, a search, a route and a purchase say “buys coffee near work on weekdays”. Change the details (a search for strollers, a visit to a baby-goods shop, a purchase of prenatal vitamins) and the same three arrows say “possibly expecting a child”: a health inference that nobody typed in.

THE COMBINATION IS THE INFERENCE

Ordinary signals become revealing when they are joined. The join is the decision to review, not only the store.

Why is this connected?

Every line between two of those boxes should be able to answer six plain questions:



A reviewer then has four answers, not one:

- **Keep the connection**, because the purpose needs it.
- **Scope it**: give the receiving service its own identifier, so two services can no longer recognise the same person.
- **Shorten retention**, so the history a pattern needs never builds up.
- **Separate the contexts**, so the join never happens.

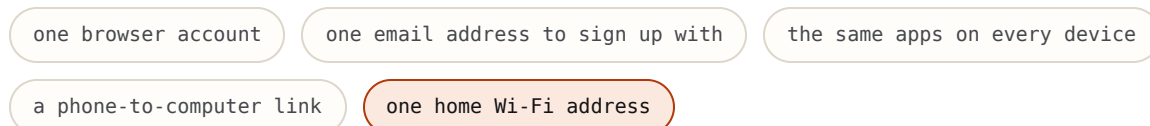
Each answer removes a particular inference and leaves the rest. Scoping the maps service's identifier does not stop it seeing a commute; it stops the commute being joined with the itinerary. Shortening the location history to a day stops the commute from forming at all.

One feature, every source

To answer “When is my flight?”, an assistant needs the question and the reservation. The calendar helps, and so does the time zone. Earlier conversations, files and location history were merely within reach. Each source is reasonable on its own. The new question is **what happens when one feature is allowed to see all of them at once**. Give each answer the sources it needs, ask before the ones that only help, and leave the rest out: the answer still works.

Many devices, many companies

Few people live inside one ecosystem. A phone from one company, a laptop from a second and a tablet from a third each sign in to their own company's account, and no single account sees every device. That does not stop the linking. It moves it to whatever runs on every device:



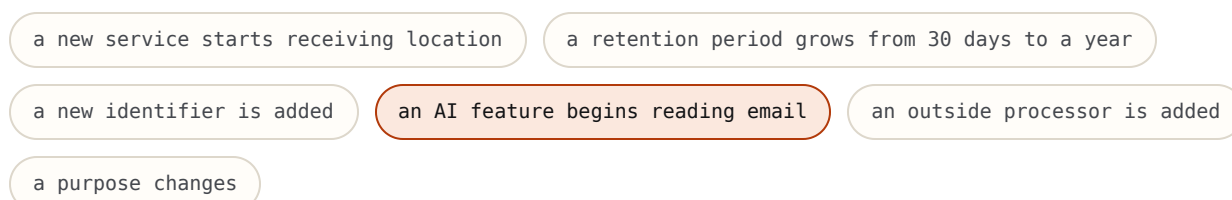
MIXING PLATFORMS MOVES THE JOIN; IT DOES NOT REMOVE IT

The question is no longer which company sees every device, but which of the things you use everywhere can recognise you everywhere.

Watches, cars and televisions join the same way: through the phone they pair with, an account, or the home network.

Privacy problems arrive as changes

Many of these arrows were not in the first design. They arrive later, one change at a time, and each deserves a review of its own:



Everyday arrows, through the four lenses

SECURITY

- Which identifiers travel with each event, and who could steal or replay them?
- Are the recovery links (phone number, recovery email, devices, IP address) kept inside security?

PRIVACY

- What can be inferred when this event is joined with others?
- Which joins does the purpose actually need?

QA

- When a connection is scoped, shortened or separated, does the inference it enabled really stop?
- Does it stop on every device and platform the person uses?

DATA GOVERNANCE

- Which connections changed since the last review, and did each change go through one?
- Who receives each derived record, and for how long?

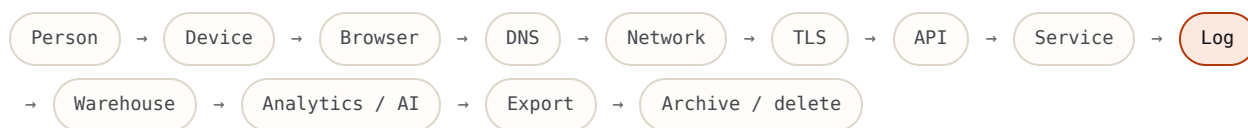
This section describes patterns, not any company's products, so it carries no citations. Try it on synthetic data in the [Privacy Command Center](#): follow "Bought coffee", ask the assistant about the flight, or open [devices across platforms](#) and change the phone, laptop and tablet.

13 EVERY LAYER

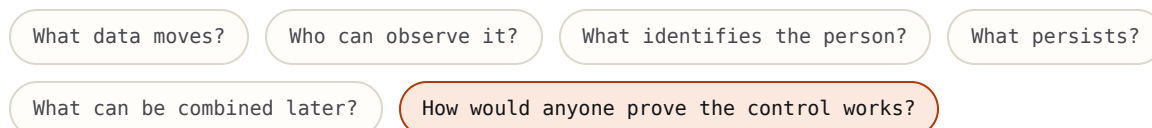
A guarantee holds until the next layer.

Every arrow above crosses layers that different specialists own: the network, the browser and device, identity, the service, the data platform, analytics and AI. Each specialist is usually right about their layer. The privacy question is where a guarantee that is true in one layer **stops being true in the next**, and the only way to find it is to follow the data across their boundaries.

Different specialists, one continuous flow of data: from the person to persistence, with the same questions at every hop. [Open full size.](#)



Every hop exposes something different: content, metadata, identity, or a new copy. So every hop gets the same questions, plus one that turns an answer into evidence:



Encryption hides the content, not the shape

Every packet wears its addresses on the outside. Transport encryption covers the inner layers; source and destination addresses, ports, packet sizes and timing stay visible to every network on the path. An IP address is not a name, but an IP address plus a precise time is a join key: inside a company's own logs it links a request to a login, and outside them a network operator's records can resolve it to a subscriber. It is probabilistic (shared Wi-Fi, NAT and VPNs blur it), which makes it linkable, not proof of identity.

Before the page even loads, a DNS lookup may already say where someone is going. Encrypted DNS hides the question from the local network and moves the trust to the resolver; it does not remove it. A resolver log looks like plumbing, but joined to the network's address leases and an asset inventory it becomes one person's browsing history, though no single table held it. Then the hostname can appear again in the encrypted connection's opening message unless Encrypted Client Hello is used, and the sizes and timing of what follows can distinguish reading from downloading. Nothing was decrypted, yet an observer can hold a sensitive fact tied to a device and a time.

THE CONTROL IS ON THE JOIN, NOT ONLY ON THE WIRE

Truncate addresses where they are collected, keep raw ones briefly, store the region when the region is the purpose, and gate the joins between network logs and the tables that name people.

TLS protects the arrow. Privacy decides whether the arrow should exist.

TLS proves the server's identity and keeps the traffic confidential and intact in transit. Its newer version also encrypts the server's certificate and makes forward secrecy mandatory, so a key stolen later cannot unlock recorded traffic. But HTTPS hides the request from the network, not from whoever terminates TLS, and that is often a CDN or a load balancer rather than the application:

WHO CAN READ WHAT, ONCE HTTPS IS ON

Element	The network path	Whoever terminates TLS	The application
Destination address	Visible	Visible	Visible
Hostname	Visible unless Encrypted Client Hello	Visible	Visible
URL path and query	Hidden	Visible	Visible
Headers, cookies, tokens	Hidden	Visible	Visible
Request body	Hidden	Visible	Visible
Sizes and timing	Visible	Visible	Visible

After termination the plaintext goes somewhere: application logs, analytics pipelines, the warehouse, backups, exports. Query strings and headers are copied into proxy, CDN and application logs by default, and logs are databases that engineers often forget are databases. Even before anything is encrypted, the opening of the connection lists the client's capabilities, which can identify the software: useful against fraud and bots, a tracking signal once it is combined with an address and a time.

From the network to identity: what each layer can observe, and where the identifier moves next. [Open full size.](#)

Identity does not require a name

Privacy problems often begin with the ability to say “this is the same browser again”, not with knowing anyone's legal name. A script embedded from another site runs with the page's own privileges; an embedded frame is isolated but still learns about the visit, so choosing between them is a data-sharing decision, not a front-end detail. And tracking is a data-flow problem rather than a cookie problem: block one mechanism and the identifier moves to another.

BLOCK ONE MECHANISM, AND THE IDENTIFIER MOVES TO THE NEXT

Mechanism	How it links you	The usual countermeasure
Third-party cookie	The same tracker cookie is seen on every site that embeds it	Blocking or partitioning
Cookie syncing	Trackers swap their identifiers through redirects	Blocking third-party cookies
Link decoration	An identifier added to the URL carries identity to the next site	Stripping known parameters

Mechanism	How it links you	The usual countermeasure
First-party disguise	The tracker is served from the site's own domain	Uncloaking; limits on script cookies
Hashed email	The same email-derived identifier on every site you sign in to	Policy and consent, not the browser

At scale, tracking is graph construction. Every event that carries two identifiers at once becomes an edge. Logins, account IDs and hashed emails are strong, exact edges; an address and a timing pattern are weak, probabilistic ones. Everything connected becomes one profile, and every system downstream inherits links it never reviewed, including wrong ones. A false merge is a harm too: one bad edge attaches someone else's behaviour to a person.

After sign-in, a session token stands in for the password on every request. Whoever presents it is treated as the user, and signed tokens are often readable by anyone who sees them, with an email address or account ID inside, so a log that contains tokens is a sensitive log. Passkeys remove the shared secret and give every site its own key, so sites gain no common identifier; the remaining risk moves to sync, recovery and what a site logs alongside the credential.

Hiding one value does not hide the pattern

A tracking pixel is just an image with a different address for each recipient; opening the email writes a row (who, when, from which address and client) and that row is joined to the customer record. Tokenisation hides the card number, but a merchant can still recognise a returning card, and one loyalty scan or e-receipt ties that reference to a named customer. Then the history speaks for itself: routines, a likely home or workplace, recurring pharmacy purchases. None of it was collected directly.

Not every sensitive signal should simply go. Fraud harms people, and a signal that measurably reduces it protects them. The better question is which signal creates the benefit, and what is the least exposing form that keeps it: a derived risk score instead of the raw trail, coarser location, a shorter retention window.

Below the API, every hop is a copy

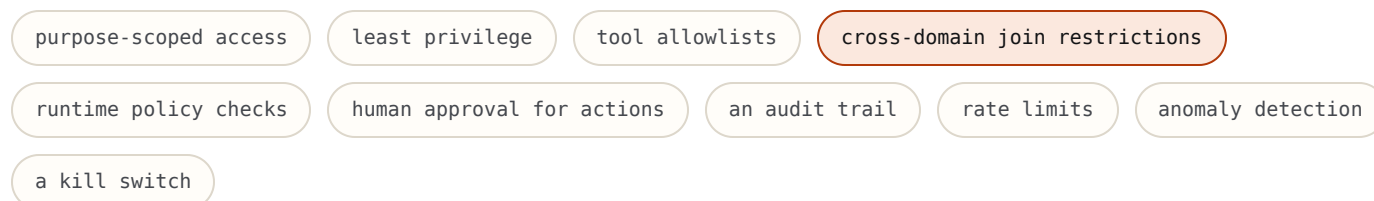
Once a request lands, the arrows become copies: a log, an event stream, object storage, the warehouse, features, a model, an export. Each copy has its own purpose, retention, access and deletion path, and the questions follow it there:

- **A join can create knowledge no table held.** Two tables can each pass review and still produce a new exposure when a shared key joins them. Review join paths, not only datasets.
- **Filtering is not isolation.** Row-level security protects the database; the semantic layer, cache, export and browser must keep the tenant's boundary too. A report that is almost entirely correct is still a complete privacy failure for the rows that are not.
- **Consent is distributed state, not a checkbox.** Every consumer holds a copy that can lag, so revocation needs a target, a check when a job reads data, a plan for work already in flight, and an acknowledgement from each consumer.
- **Delete, recover and preserve are different states** of every copy, which is [the arrow into the future](#).

An agent can create the join that nobody wrote

An AI agent is a loop: the model reads its context, calls a tool, and reads the result as new input, sometimes saving facts to memory for later. Retrieved documents and tool results are data and possible instructions at the same time,

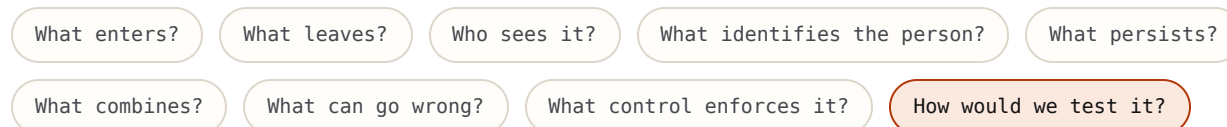
and similarity search finds related text, not permitted text. Authorised inputs do not automatically create an authorised inference. Since the model is not a policy engine, the invariants have to hold in the runtime around it, even if the model itself is manipulated:



The whole map, from packet to model: every surface, and the questions that travel with the data. [Open full size.](#)

When the system is unfamiliar, ask the same questions

Nobody knows every implementation. Reasoning correctly about the data does not require it:



EVERY GUARANTEE HAS A BOUNDARY

A control can be correct in one layer and quietly stop being true in the next. The work is to follow the data across that boundary, and to prove it holds on the other side.

This section describes how common protocols and architectures generally behave, not any company's products, so it carries no citations. Details vary by version, configuration and platform; check them before relying on one.

14 WHAT THE COMPARISON TEACHES

Ten patterns across eight products.

Read the tables side by side and the same few arrows keep deciding the outcome, whichever company built the product.

01 The recovery path is the real boundary.

Passkeys, messages and device backups are end-to-end encrypted by all three companies that offer them. What differs is who can restore the key when every device is gone: a recovery contact, a PIN, the screen lock, or the provider. Whoever can restore it is inside the boundary.

Seen in [sign-in](#), [messages](#) and [cloud backup](#).

02 Encryption answers who can read it, not why it exists.

Every mail provider here encrypts stored mail; none end-to-end encrypts the subject line of ordinary email. WhatsApp encrypts every message and still documents collecting who you talk to and when. Encryption is a security answer to a privacy question it does not ask.

Seen in [mail](#) and [messages](#).

03 The default is the product.

Safari limits known fingerprinting scripts in every tab; Chrome keeps third-party cookies on. WhatsApp's encryption cannot be switched off; Apple's strongest backup protection must be switched on. Google keeps Gemini activity unless you turn it off. Most people never change a default, so the default is what the product does.

Seen in [browser](#), [messages](#), [cloud backup](#) and [AI assistant](#).

04 Metadata is data.

A tracking pixel reveals when and where you read. Message servers keep lookup logs and connection details. Wallets keep transaction details, or document that they do not. The content can be sealed while the pattern around it stays readable.

Seen in [mail](#), [messages](#) and [wallet](#).

05 A claim you can verify is a control. One you cannot is a promise.

Apple publishes measurements of the software its private cloud runs; Proton publishes its apps' source and audits; Google published its age-proof cryptography; WhatsApp publishes key transparency. Everywhere else the evidence is a bug bounty and a policy page. QA's job is to turn each promise into a test someone can run.

Seen in [AI assistant](#), [mail](#), [wallet](#) and [messages](#).

06 Deletion has to follow every copy.

A passkey deleted on the phone can live on at the site. A Gemini chat reviewed by a person is kept for up to three years after you delete your activity. A voice recording can be deleted while its transcript survives. Every 30-day bin is a copy with a clock. Governance is counting the copies.

Seen in [sign-in](#), [AI assistant](#), [home & voice](#) and [cloud backup](#).

07 A setting can be withdrawn.

Amazon removed a local-processing option from some Echo devices in 2025. Apple stopped offering Advanced Data Protection to new users in the United Kingdom. Google retired most of the technology it had built to replace third-party cookies. A control that the vendor or a government can remove needs a date on it, and a test that notices when it goes.

Seen in [home & voice](#), [cloud backup](#) and [browser](#).

08 Recoverability is a design decision.

Encryption, tokenization, hashing, keyed pseudonyms and deletion are not interchangeable. Some keep the original recoverable for whoever holds a key or a vault. Some keep only the ability to match. Some destroy recoverability on purpose. Choosing between them is not a security detail to settle later: it fixes, years in advance, what the organisation can recover, search, produce or prove, including under a legal hold. So the future obligations have to be decided before the transformation is chosen, not discovered after the key is gone. The privacy question is not simply whether data is protected today. It is what the system must still be able, or unable, to know tomorrow.

Seen in [the arrow into the future](#), [cloud backup](#) and [sign-in](#).

09 The link moves; it does not disappear.

A scoped identifier, a separate account or a device from another company each removes one join. The person is then joined again by whatever is shared: an email address, a browser account, an app, a home network, an assistant allowed to read everything. Review the joins as carefully as the stores: a combination of ordinary records is new information that nobody typed in.

Seen in [browser](#), [AI assistant](#) and [everyday arrows](#).

10 Every guarantee has a boundary.

TLS protects the arrow, not what happens after it. Row-level security filters the database, not the export. A passkey removes the password, not the recovery path. A token hides the card, not the purchase history. Each control is correct in its layer, and each stops at a boundary that someone else owns. The finding is usually not a broken control but a correct one trusted one layer too far.

Seen in every layer, sign-in, wallet and mail.

15 THE FOUR-LENS REVIEW

Twenty-eight questions for any product.

Use them on a product you are building, buying or trusting with your family's data. A question nobody can answer is a finding.

SECURITY

- What stops a look-alike site, a stolen session or a replayed request?
- Where does the key live, and what hardware protects it?
- Who can restore access when every device is lost, and how are guesses limited?
- What is sent before the person acts: before the wake word, before the tap, before opening?
- What happens on a compromised or second-hand device?
- If recovery is required, who controls the key or the mapping?
- Are the links kept for security (phone number, recovery email, devices, IP address) used for nothing else?

PRIVACY

- What leaves the device, and could it have been computed on the device?
- Which identifier links this activity to other activity, and is it scoped?
- What metadata remains after the content is encrypted?
- Is it used for training, ads or personalisation, and what is the default?
- Could the purpose be met by proving an attribute instead of revealing the data?
- Should the original remain recoverable, or would a pseudonymous or irreversible representation meet the purpose?
- What can be inferred when this activity is joined with other activity, on this device and on the person's other devices?

QA

- What would an outsider need to verify the claim: source, logs, an audit, a research environment?
- Is the default state tested, not only the configured one?
- Is every fallback tested: to SMS, to an unencrypted channel, to a password?
- Is deletion tested end to end, including derived data and backups?
- Is the test re-run after every release and every policy change?
- Can we restore and interpret an old protected record using the documented recovery path?
- When a connection is scoped, shortened or separated, does the inference it enabled really stop?

DATA GOVERNANCE

- Who holds the key, and who could be compelled to use it?
- How long does each copy live: active, backup, reviewed, derived?
- Can the person export it, and in what format?
- Does deletion in one place reach every other place?
- Which controls can the vendor or a government withdraw, and who would be told?
- Does a preservation hold include the keys, mappings, schemas, metadata and derived copies needed to make the record meaningful?
- Which connections changed since the last review (a new receiver, identifier, processor, purpose or retention period), and did each go through review?

[Download the review \(Markdown\)](#)

16 HOW THIS WAS MADE

What was checked, and what was not.

Every claim in the tables comes from the company's own documentation: support pages, security guides, legal notices, newsroom and engineering posts. Four exceptions are labelled in the sources: an IETF presentation, a published third-party audit, a regulator's press release and one news report of a dated change. Sources were reviewed on 30 September 2026. Several vendor pages could be confirmed only through their search listings, so the wording here is a paraphrase, not a quotation; check the current page before you rely on one.

The companies are compared on what they document, not on how well they meet it. A documented limitation is not a failure, and a missing row is not a strength. The Test entries are recommendations, not findings: none of them has been run for this essay.

What could not be confirmed

- The default of Apple's Protect Mail Activity setting, and whether Safari's broader tracking-protection setting is on outside Private Browsing.
- Whether Gmail's image proxy hides the reader's IP address, in Google's own words.
- Whether Firefox's newest fingerprinting defences are now on in Standard mode.
- The default of Google Wallet's personalisation setting.
- Whether Nest camera video is end-to-end encrypted, and whether Private AI Compute has a public transparency log.
- The current default of Amazon's setting for human review of voice recordings.

Corrections are welcome, with a link to the company's page: [contact](#).

Earlier editions

Edition 3 followed one fictional person, Dana, through a fictional company and her connected home in thirty scenes. It is kept as it was, with every scene and link: [read edition 3](#). The same fictional company now lives in the [Privacy Command Center](#), where each product section above links to the matching view on synthetic data.

Every product is a bundle of arrows. Security, privacy, QA and data governance are four ways of asking whether each one was decided on purpose.

A person does not live inside one application. They sign in, browse, write, pay, back up, ask and speak, through products built by different companies with different defaults, on devices from different companies. Privacy engineering must follow them across all of those boundaries.

Some arrows move information through space. Some move it through systems. And some decisions determine whether information can still be recovered years later.

Some arrows copy data.
Some arrows create an identity.
Some arrows make an inference.
And some arrows open the door.
Every arrow is still a decision.

Sources

Each source is also cited beside the claim it supports. Grouped by company; the last group holds the standards body, auditor, regulator and press report.

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